

P-hacking with Random Data Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 35-45 min

Category	Natural Sciences	Time	35-45 min
Difficulty	Extended	ID	p-hacking-with-random-data

Big idea: Methods can create false confidence.

One-Minute Teacher Brief

Students experience how false patterns can appear when many tests are tried.

Student Prompt

Inspect P-hacking with Random Data Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

P-hacking with Random Data Stimulus A	Student-facing stimulus 1: A random dataset where students hunt for any impressive correlation. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
P-hacking with Random Data Stimulus B	Student-facing stimulus 2: Multiple subgroup comparisons that eventually produce a 'significant' pattern. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
P-hacking with Random Data Reveal / Twist	Student-facing stimulus 3: A publication pitch followed by a methods audit. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Methods can create false confidence.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting P-hacking with Random Data Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from P-hacking with Random Data Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Can correct methods lead to false knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how methods can create false confidence.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Can correct methods lead to false knowledge?

Setup Checklist

- Prepare a starter stimulus using: A random dataset where students hunt for any impressive correlation.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A random dataset where students hunt for any impressive correlation.
- Multiple subgroup comparisons that eventually produce a 'significant' pattern.
- A publication pitch followed by a methods audit.

Resources To Use

- Resource pack: <resources/p-hacking-with-random-data/index.html>
- Card deck CSV: <resources/p-hacking-with-random-data/cards.csv>
- Visual prompt: <resources/p-hacking-with-random-data/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give groups a random data table.
2. Let them test as many relationships as possible.
3. Reward groups for finding an apparently interesting pattern.
4. Reveal that, with enough searches, chance findings become likely.

Debrief Moves

- Knowledge question: Can correct methods lead to false knowledge?
- Why are preregistration, replication and transparency important?
- How can a result be statistically impressive but epistemically weak?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...